3 Yr. Degree/4 Yr. Honours 4th Semester Examination, 2025 (CCFUP)

Subject : Computer Science

Course: COMP4013 (MAJOR)

Database Management Systems

Time: 2 Hours **Full Marks: 40**

The figures in the right hand margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* questions: 2×5=10

(a) What is Functional dependency?

(b) Define candidate key.

(c) What do you mean by cardinality of a relation?

(d) What is the difference between Schema and Instance?

(e) What is data independence?

(f) What is a view?

(g) Define weak entity.

(h) What is Natural Join?

2. Answer *any two* questions: 5×2=10

(a) Describe the three levels of data abstraction in a DBMS. What is meta data? 4+1

(b) Explain the role and functions of the database administratior. Define instance. 4+1

(c) Briefly describe ACID properties in transactions. 5

(d) What is Serializability in transaction management? Explain Two-Phase Locking Protocol (2PL). 2+3

3. Answer *any two* questions: 10×2=20

(a) Define DBMS. Explain its advantages over a traditional file system. Explain Referential Integrity Constraint with an example. 2+5+3

(b) What is Normalization? Explain 1NF, 2NF and 3NF briefly with examples. 2+8

(c) What is ER Model? Draw an ER diagram for a Hospital Management System. 2+8

(d) Write short notes on (*any two*): 5×2

(i) Index Sequential File Organization.

(ii) Data Dictionary

(iii) Relational Algebra

(iv) Multilevel indexing using B+ tree